

The Data Design Processes

DATA 3302: Data Visualization

Professor Austin P. Wright

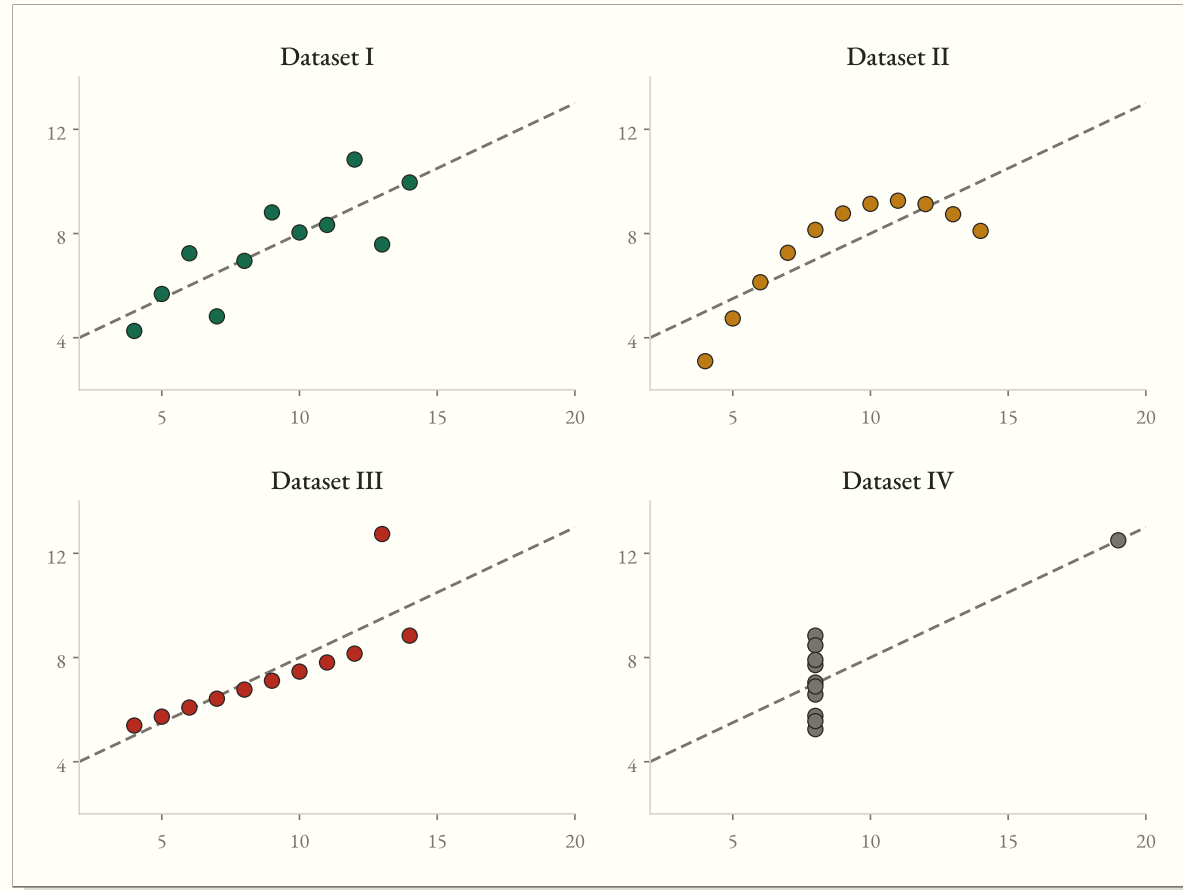
Cal Poly San Luis Obispo · Fall 2026

Slides based on Munzner, Visualization Analysis and Design, 2014

Why involve direct representations?

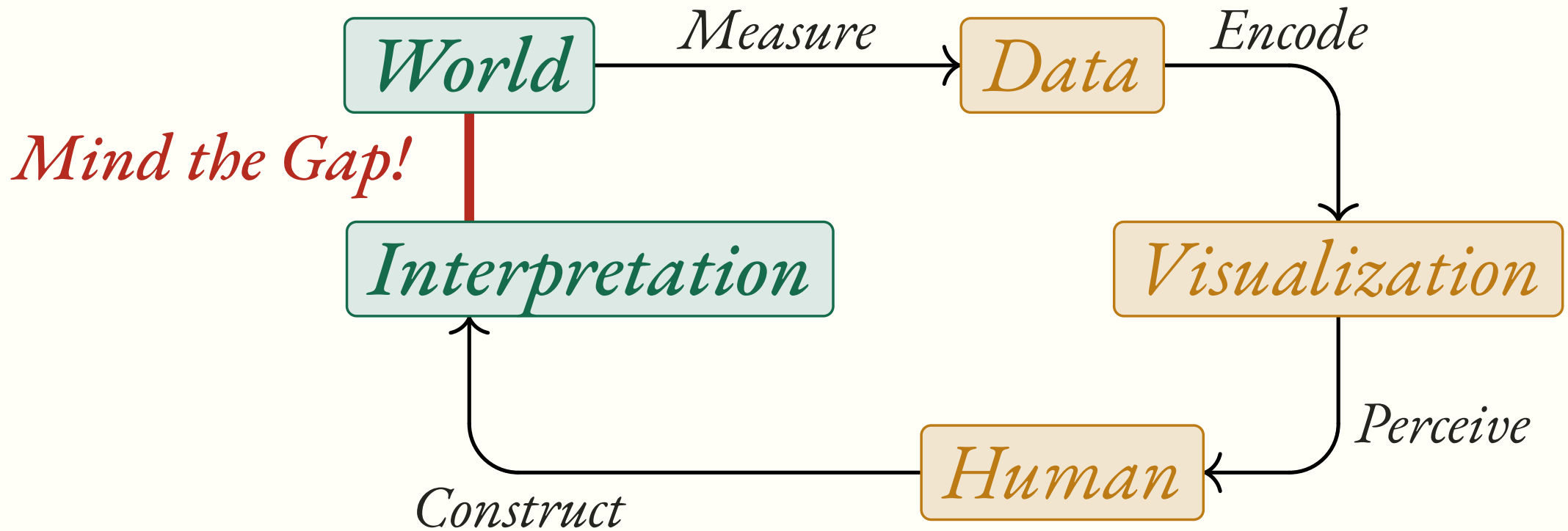
Anscombe's Quartet

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
Mean of x	9.00	9.00	9.00	9.00
Mean of y	7.50	7.50	7.50	7.50
Variance of x	11.00	11.00	11.00	11.00
Variance of y	4.13	4.13	4.12	4.12
Correlation	0.82	0.82	0.82	0.82
Regression Fit	$y = 3 + 0.5x$	$y = 3 + 0.x$	$y = 3 + 0.5x$	$y = 3 + 0.5x$



Anscombe, "Graphs in Statistical Analysis," 1973

Why is this difficult?



Data Abstraction

Conceptual Model vs Data Model

Data Types

1. Items / Hard Index
2. Attributes / Variables
3. Links / Relations
4. Positions / Soft Index
5. Grids / Topologies

Data Types

1. Items / Hard Index
2. Attributes / Variables
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Dataset Types

1. Tables
2. Networks
3. Fields
4. Geometry
5. Sets

Dataset Availability

1. Static / Offline
2. Dynamic / Online

Dataset Mutability

1. Fixed / Immutable
2. Reactive / Mutable

A City Bike-Share System

Riders check out *bikes* from *stations* scattered across downtown. Every *trip* links a start station to an end station. Each station reports its *GPS coordinates*.

Your task is to build a continuously updated citywide *demand heatmap*. You start by identifying the data abstractions.

Identify the Types

1. Dataset Type?
2. Data Types?
3. Dataset Availability/Mutability?

Attribute Types

Attribute Types

1. Categorical / Nominal
2. Ordinal
3. Quantitative (Interval)
4. Quantitative (Ratio)

Attribute Types

1. Categorical / Nominal
2. Ordinal
3. Quantitative (Interval)
4. Quantitative (Ratio)

Order Directions

1. Sequential
2. Diverging
3. Cyclic

A	B	C	S	T	U
Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
32	7/16/07	2-High	Small Pack	0.79	7/17/07
32	7/16/07	2-High	Jumbo Box	0.72	7/17/07
32	7/16/07	2-High	Medium Box	0.6	7/18/07
32	7/16/07	2-High	Medium Box	0.65	7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
66	1/20/05	5-Low	Wrap Bag	0.56	1/20/05

Example Dataset of Orders with Attribute Types

Data Semantics

Abstract Semantics

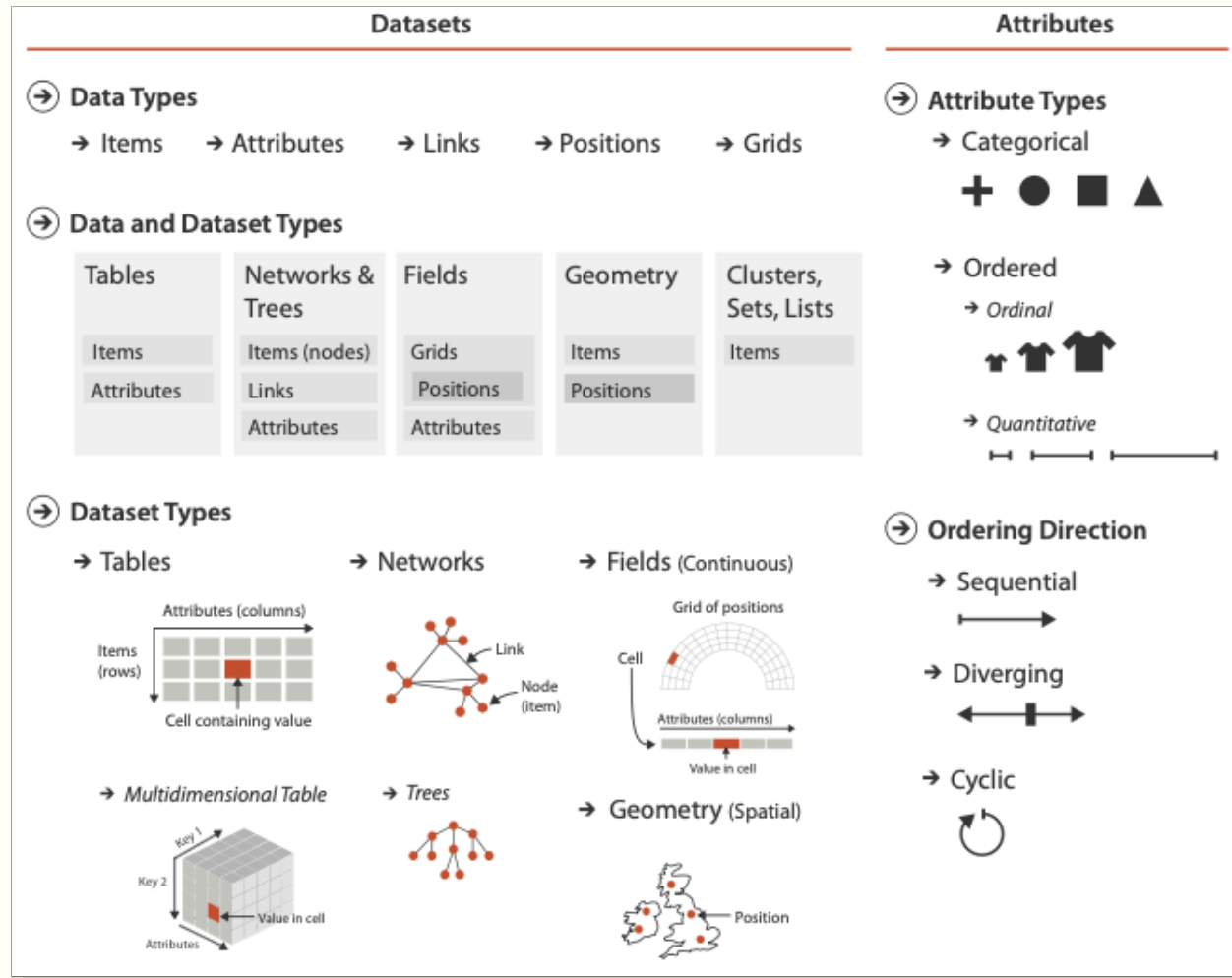
1. Key / Index
2. Value / Measure

Abstract Semantics

1. Key / Index
2. Value / Measure

Structural Semantics

1. Spatial
2. Temporal
3. Field
4. Vector



Data Transformations and Tidy Data

*Data is rarely ready to visualize right away,
you'll often need to **transform**, **merge**, or **tidy** it
first.*

The Relational Model

Relational Data

1. *Table*: Dataset
2. *Row*: Record
3. *Column*: variable (name + type)
4. *Schema*: Table's column names & types
5. *Database*: Set of tables

<i>Team</i>	<i>Wins</i>	<i>Losses</i>
Warriors	10	4
Kings	7	7
Lakers	3	11

Each *row* is a record, each *column* is a variable, each *cell* is a value.

Relational Algebra

Table(s) in, table out.

Core Operations

1. *select* — choose a set of columns
2. *filter* — remove unwanted rows
3. *order by* — sort records by column values
4. *group by* + aggregate (sum, mean, ...) — partition, then summarize
5. *join / union* — combine multiple tables

Implemented by SQL, pandas, polars, *arquero*, and similar table libraries.

Select & Filter

<i>A</i>	<i>B</i>	<i>C</i>
x	1	4
y	2	5
z	3	6

select "A", "B"

<i>A</i>	<i>B</i>
x	1
y	2
z	3

filter "B" % 2 = 1

<i>A</i>	<i>B</i>	<i>C</i>
x	1	4
z	3	6

Group By + Aggregate

group by “A”; aggregate mean

Original			→ Aggregated		
<i>A</i>	<i>B</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>C</i>
x	1	4	x	5.33	7.67
y	2	5	y	6.5	8.5
z	3	6	z	3	6
x	7	9			
x	8	10			
y	11	12			

Combining Tables: Join

inner join T_1, T_2 on “A”

T_1			T_2			$\longrightarrow$						Result					
<i>A</i>	<i>B</i>	<i>C</i>	<i>A</i>	<i>D</i>	<i>E</i>							<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	
x	1	4	x	7	10							x	1	4	7	10	
y	2	5	why	8	11							z	3	6	9	12	
z	3	6	z	9	12												

An *inner join* keeps only matching keys. An *outer join* keeps every key filling in missing values.

Based on this, what would a *left join* keep?



Tidy Data

Tidy Data

Rules of Tidy Data

1. Each *observation* is a row
2. Each *variable* is a column
3. Each *value* is a cell

<i>month</i>	<i>temp</i>	<i>precip</i>	<i>wind</i>
January	56.55	0.06	5.32
February	56.10	0.12	5.46
March	56.36	0.11	6.52

Tidy data relies on *both* a conceptual model (what's counts as an observation?) and a data model.

Wide vs Narrow Tables

Better for *data entry* vs *data analysis*

Tidy

<i>month</i>	<i>temp</i>	<i>precip</i>	<i>wind</i>
January	56.55	0.06	5.32
February	56.10	0.12	5.46

Untidy

<i>month</i>	<i>element</i>	<i>value</i>
January	temp	56.55
January	precip	0.06
January	wind	5.32
February	temp	56.10
February	precip	0.12
February	wind	5.46

In the untidy table, a single observation is spread across several rows, and the *value* column mixes different variables. Reshaping between these forms is called *pivoting* and *unpivoting*.

Discussion

Is this table tidy? What are its variables and its observations?

<i>artist</i>	<i>track</i>	<i>date_entered</i>	<i>wk1</i>	<i>wk2</i>	<i>wk3</i>
2 Pac	Baby Don't Cry	2000-02-26	87	82	72
3 Doors Down	Kryptonite	2000-04-08	81	70	68
98°	Give Me Just One Night	2000-08-19	51	39	34

Weekly Billboard ranks, 2000 · *R for Data Science*, Hadley Wickham

One person's tidy is another person's mess

*This means that most real analyses will require at least a little tidying. You'll begin by figuring out what the underlying **variables** and **observations** are. Sometimes this is easy; other times you'll need to consult with the people who originally generated the data.*

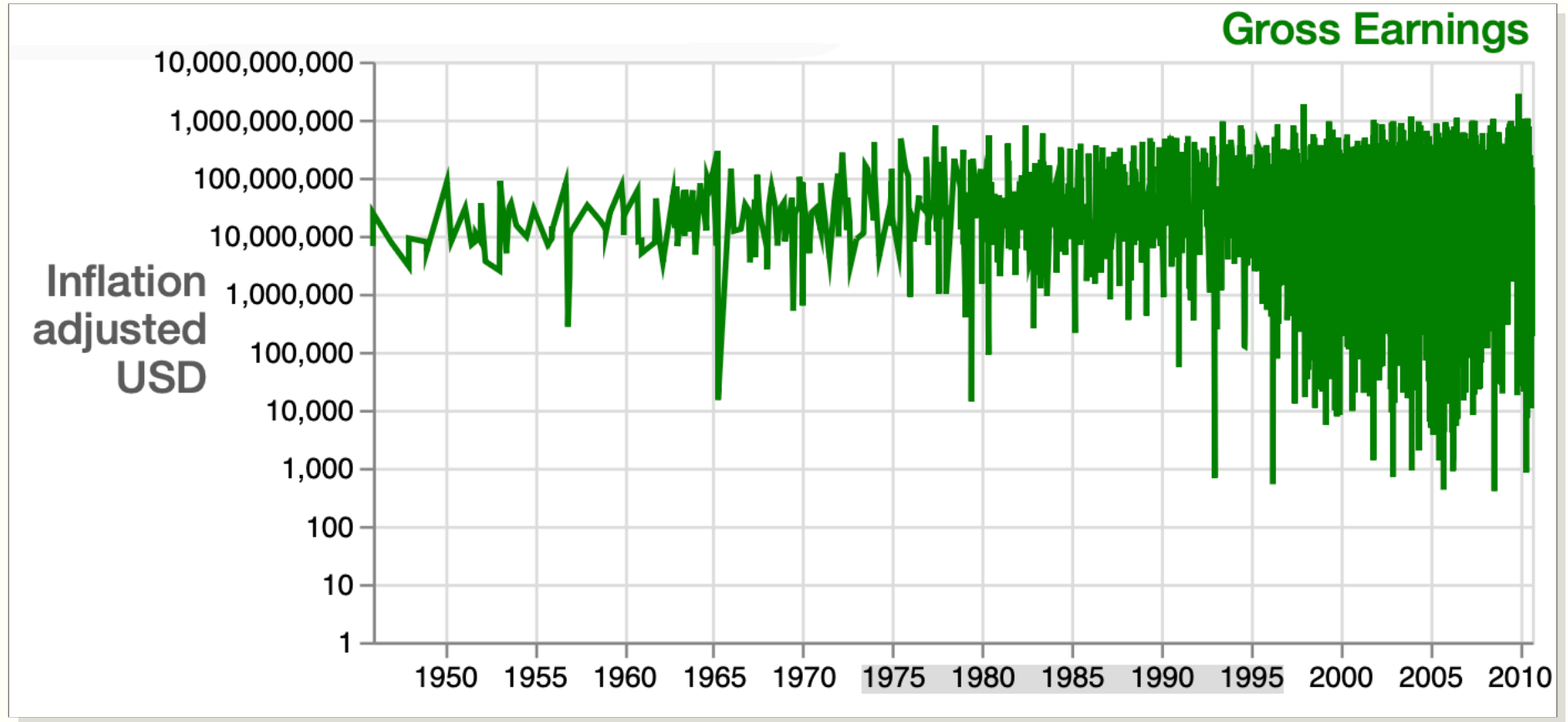
— Hadley Wickham

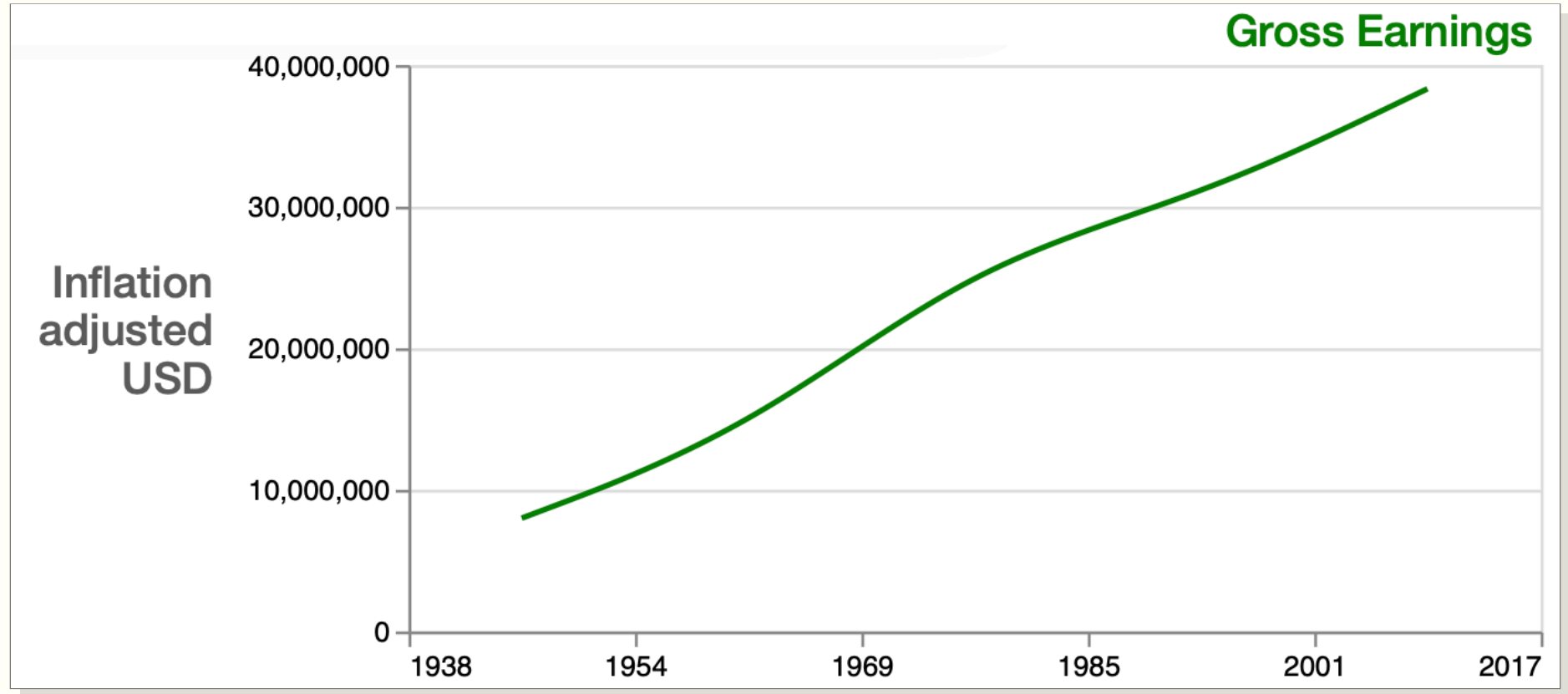
R for Data Science, “Data Tidying”

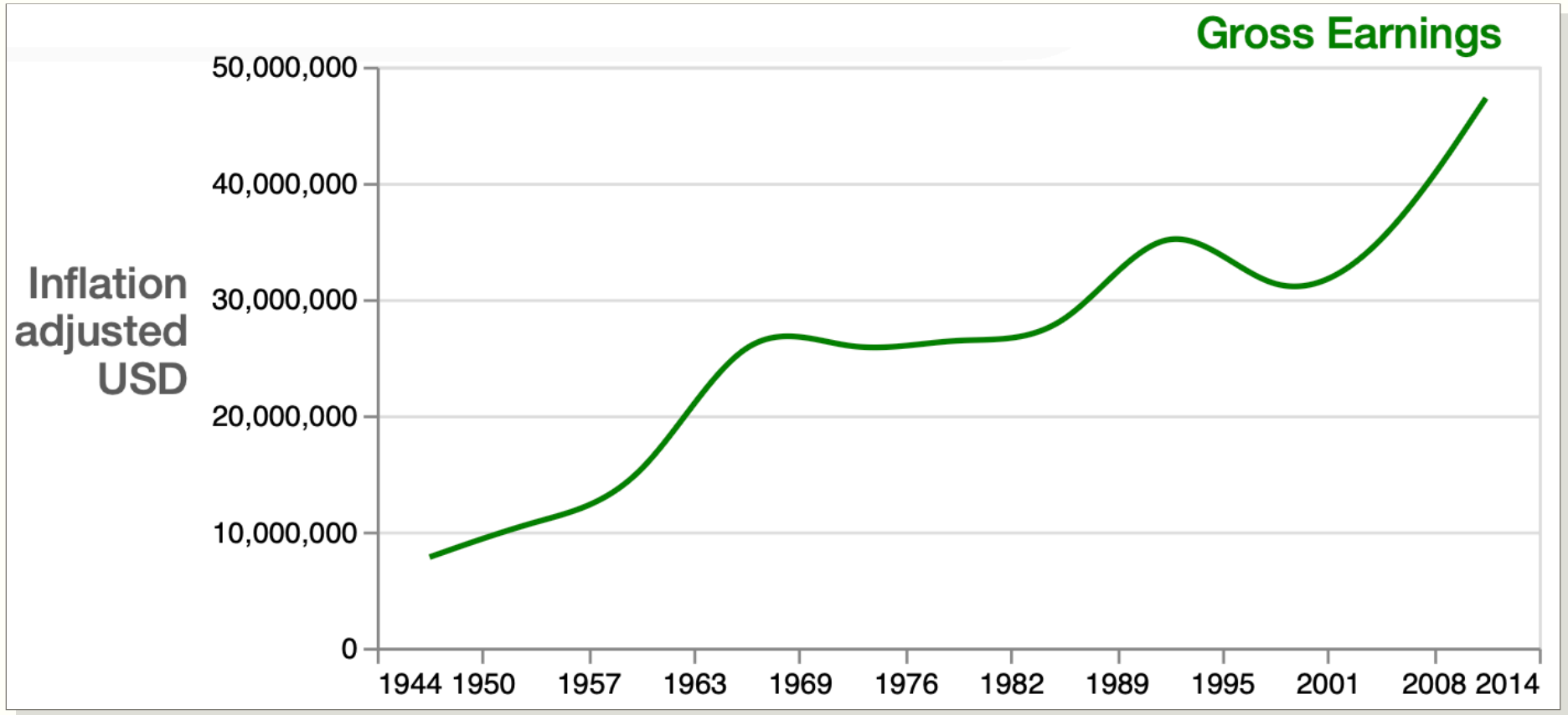


Visualization (In)Dependent Transformations

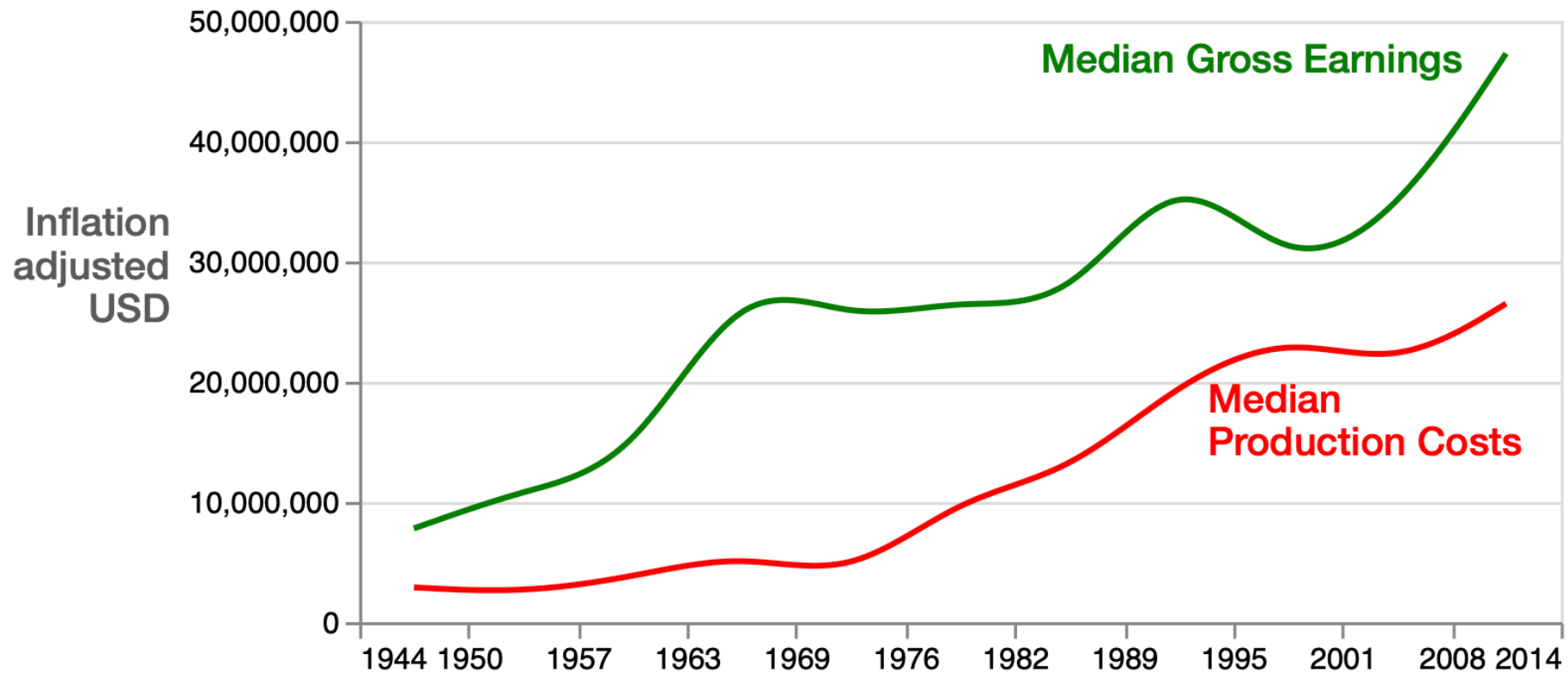
Title	Release_Date	Worldwide_Gross	Production_Budget	IMDB_Rating
The Land Girls	Jun 12 1998	146083.0	8000000.0	6.1
First Love, Last Rites	Aug 07 1998	10876.0	300000.0	6.9
I Married a Strange Person	Aug 28 1998	203134.0	250000.0	6.8
Let's Talk About Sex	Sep 11 1998	373615.0	300000.0	NaN
Slam	Oct 09 1998	1087521.0	1000000.0	3.4
...	...	...	...	...
Zack and Miri Make a Porno	Oct 31 2008	36851125.0	24000000.0	7.0
Zodiac	Mar 02 2007	83080084.0	85000000.0	NaN
Zoom	Aug 11 2006	12506188.0	35000000.0	3.4
The Legend of Zorro	Oct 28 2005	141475336.0	80000000.0	5.7
The Mask of Zorro	Jul 17 1998	233700000.0	65000000.0	6.7



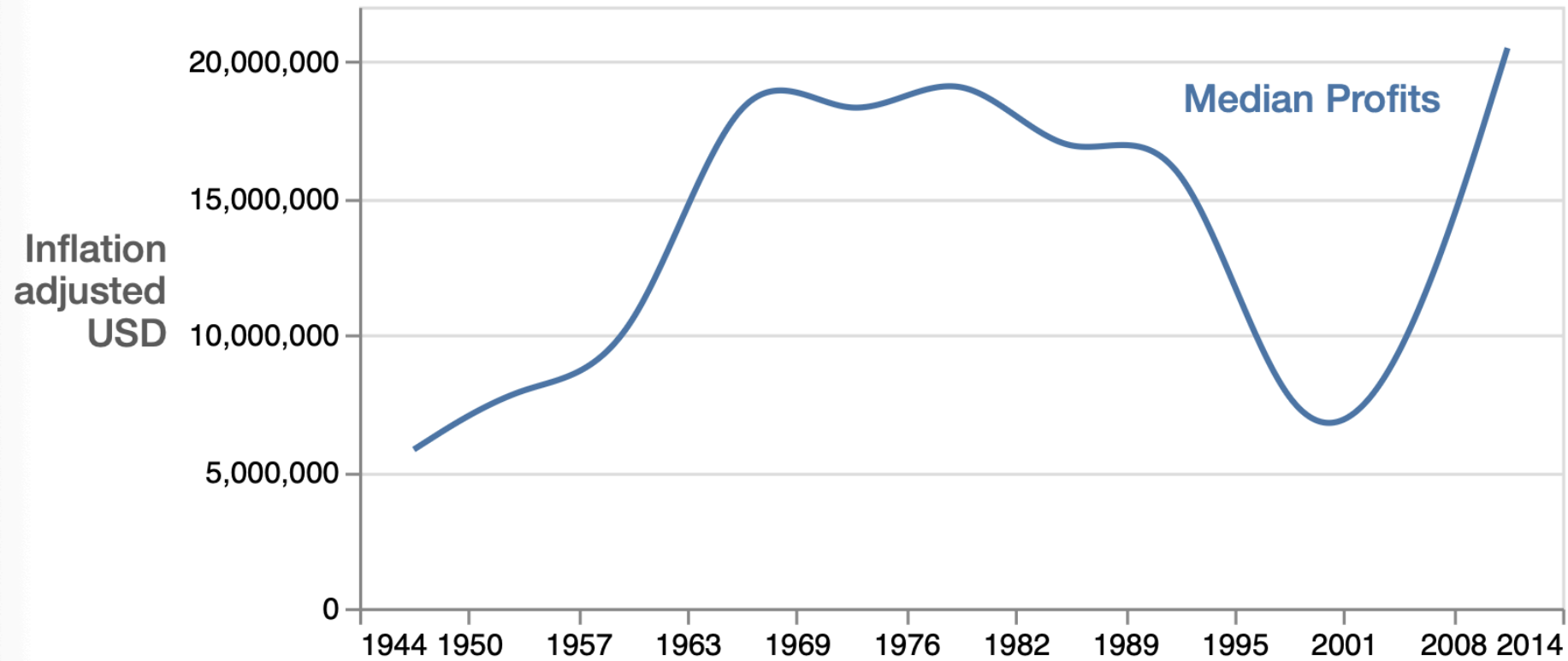




Movie Business Growing Over Time



Movie Business Growing Over Time?





Task Abstraction

*What does the **user actually do** with the
visualization?*

User Actions

Analyze

1. Consume
 1. Discover
 2. Present
 3. Enjoy

User Actions

Analyze

1. Consume
 1. Discover
 2. Present
 3. Enjoy
2. Produce
 1. Annotate
 2. Record
 3. Derive

User Actions

Analyze

1. Consume
 1. Discover
 2. Present
 3. Enjoy
2. Produce
 1. Annotate
 2. Record
 3. Derive

Search

1. Target Known
 1. Lookup
 2. Locate
2. Target Unknown
 1. Browse
 2. Explore

User Actions

Analyze

1. Consume
 1. Discover
 2. Present
 3. Enjoy
2. Produce
 1. Annotate
 2. Record
 3. Derive

Search

1. Target Known
 1. Lookup
 2. Locate
2. Target Unknown
 1. Browse
 2. Explore

Query

1. Identify
2. Compare
3. Summarize

User Actions (verbs) vs User Targets (nouns)

User Targets

Full Dataset

1. Trends
2. Features
3. Outliers / Anomalies

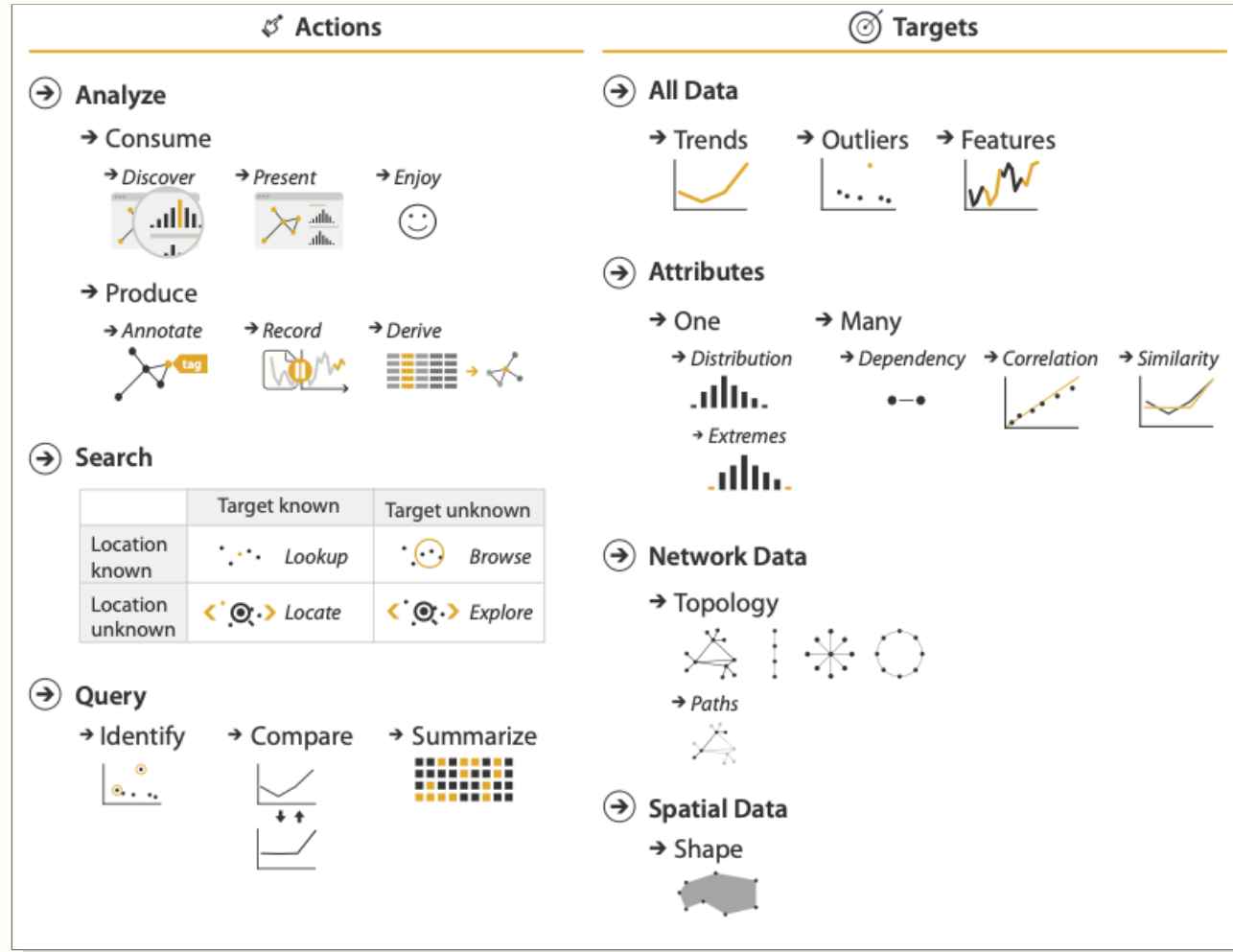
User Targets

Full Dataset

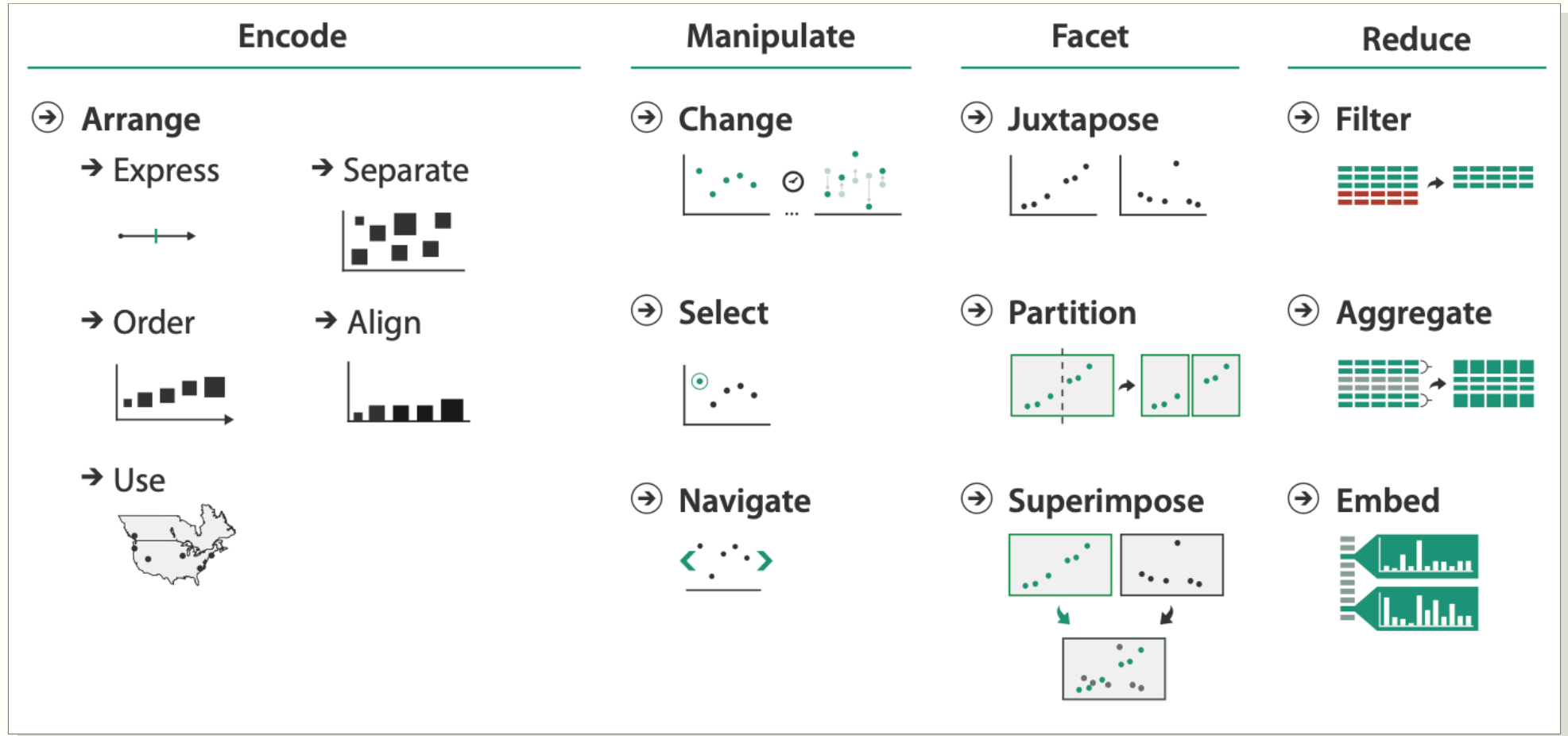
1. Trends
2. Features
3. Outliers / Anomalies

Attributes

1. Single Attribute
 1. Distribution
 2. Paradigms
 3. Extremes
2. Multiple Attributes
 1. Dependency
 2. Correlation
 3. Similarity



How do designs support user tasks?

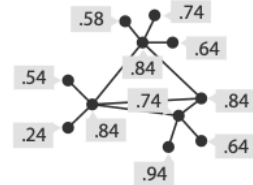


Munzner 2014, Figure 3.7

Task 1



In
Tree



Out
Quantitative
attribute on nodes



What?

- **In** Tree
- **Out** Quantitative attribute on nodes

Why?

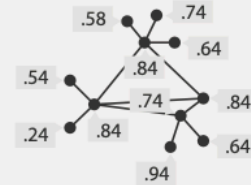
- Derive

Task 2



In
Tree

+



In
Quantitative
attribute on nodes



Out
Filtered tree
Removed
unimportant parts

What?

- **In** Tree
- **In** Quantitative attribute on nodes
- **Out** Filtered tree

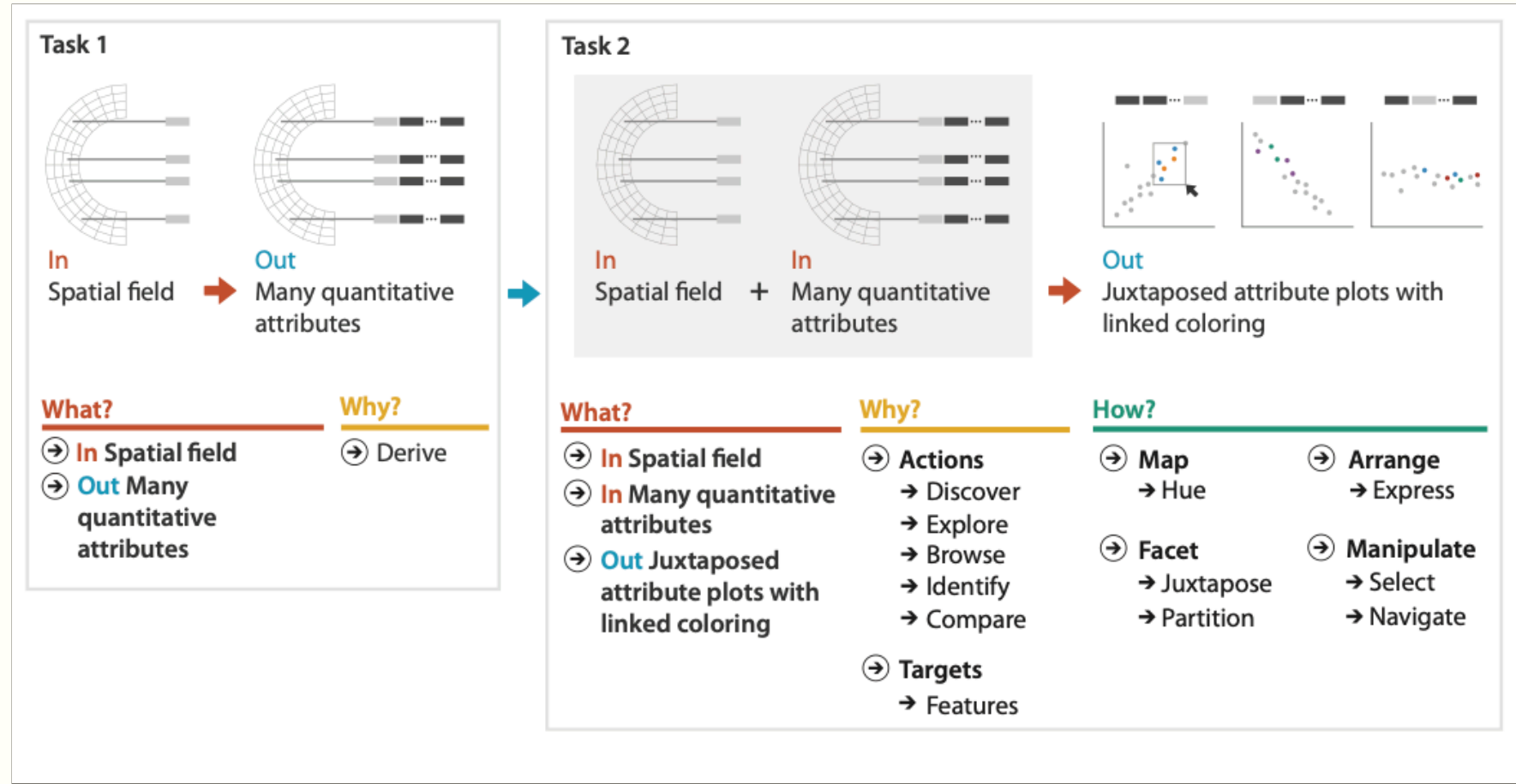
Why?

- Summarize
- Topology

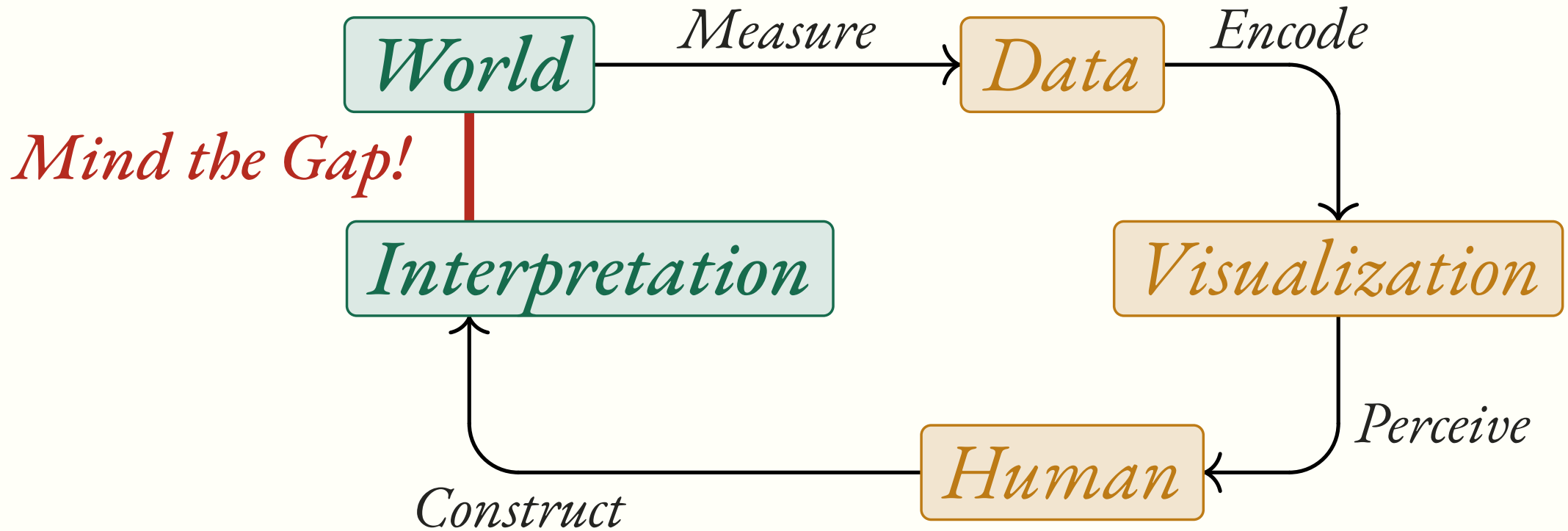
How?

- Reduce
- Filter

Munzner 2014, Figure 3.11



Munzner 2014, Figure 3.13



How do we know if we did a good job?



The Nested Model



Domain situation

Observe target users using existing tools



Data/task abstraction



Visual encoding/interaction idiom

Justify design with respect to alternatives



Algorithm

Measure system time/memory

Analyze computational complexity

Analyze results qualitatively

Measure human time with lab experiment (*lab study*)

Observe target users after deployment (*field study*)

Measure adoption

Munzner 2014, Figure 4.1

*How do we **actually make** good designs?*

The Design Process

Based on slides from Tal Wolman

*The design space of possible visualizations is huge, and includes considerations of both **how to create** and **how to interact** with visual representations. Vis design is full of trade-offs, and most possibilities in the design space **are ineffective** for a particular task — designers must weigh the resource limitations of computers, of humans, and of displays.*

— Tamara Munzner

Visualization Analysis & Design, 2014

Every task has countless possible designs, most of them are bad, and you can't tell which from a distance.

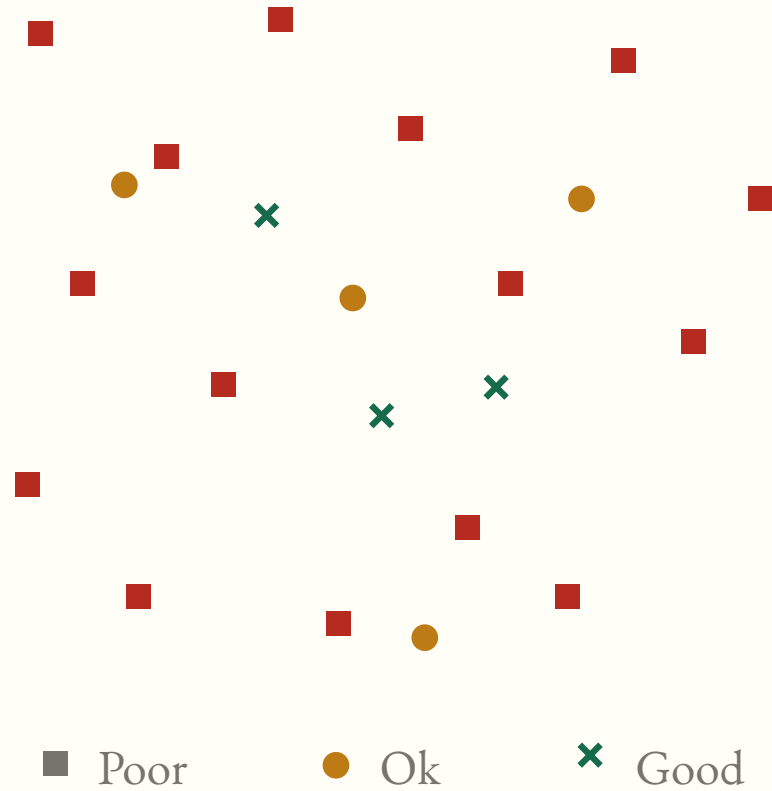
Space of Possible Solutions



Space of Possible Solutions



Space of Possible Solutions

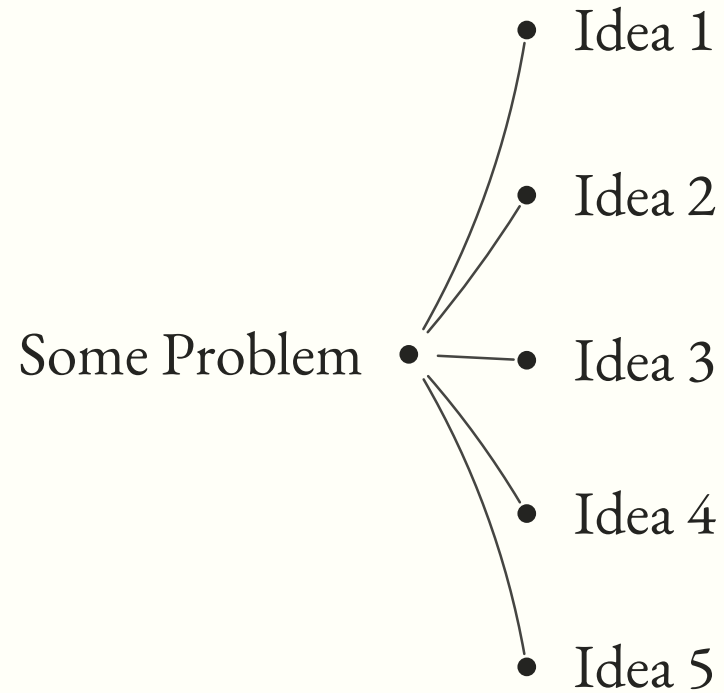


*The problem of a small consideration space is the higher probability of settling for an OK solution and missing a good one. A fundamental principle of design is to **consider multiple alternatives** and then choose the best, rather than fixating on the first idea. One way to guarantee this: **explicitly generate multiple ideas in parallel.***

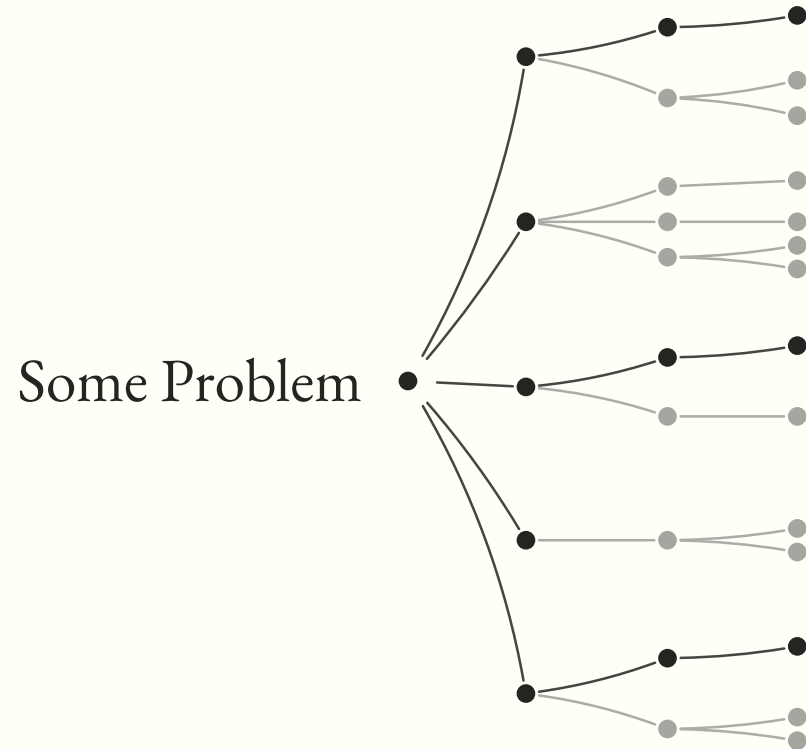
— *Tamara Munzner*

Visualization Analysis & Design, 2014

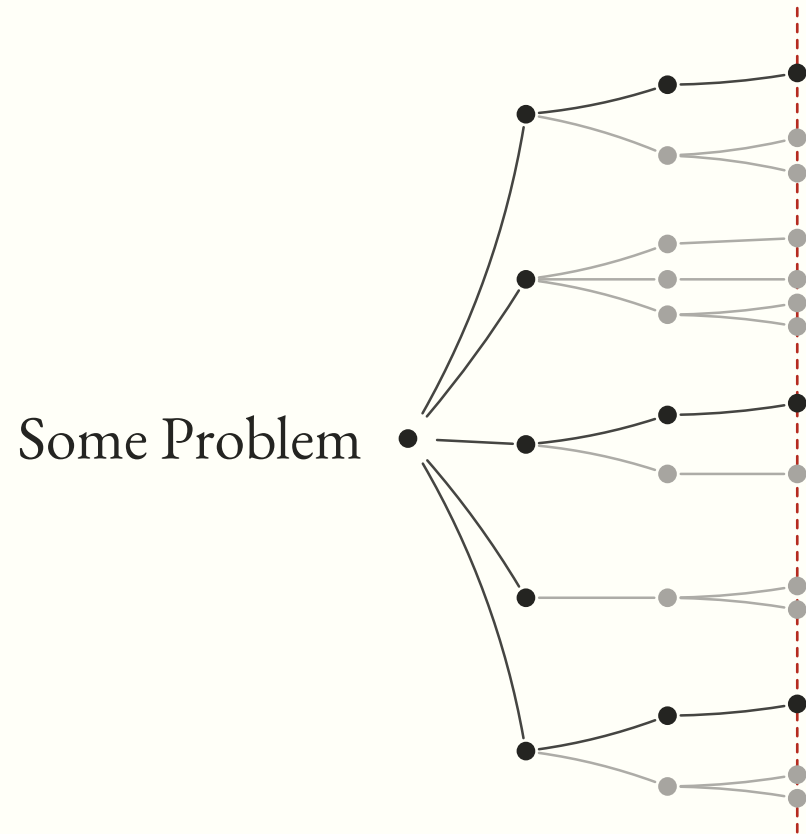
Ideation as a Tree



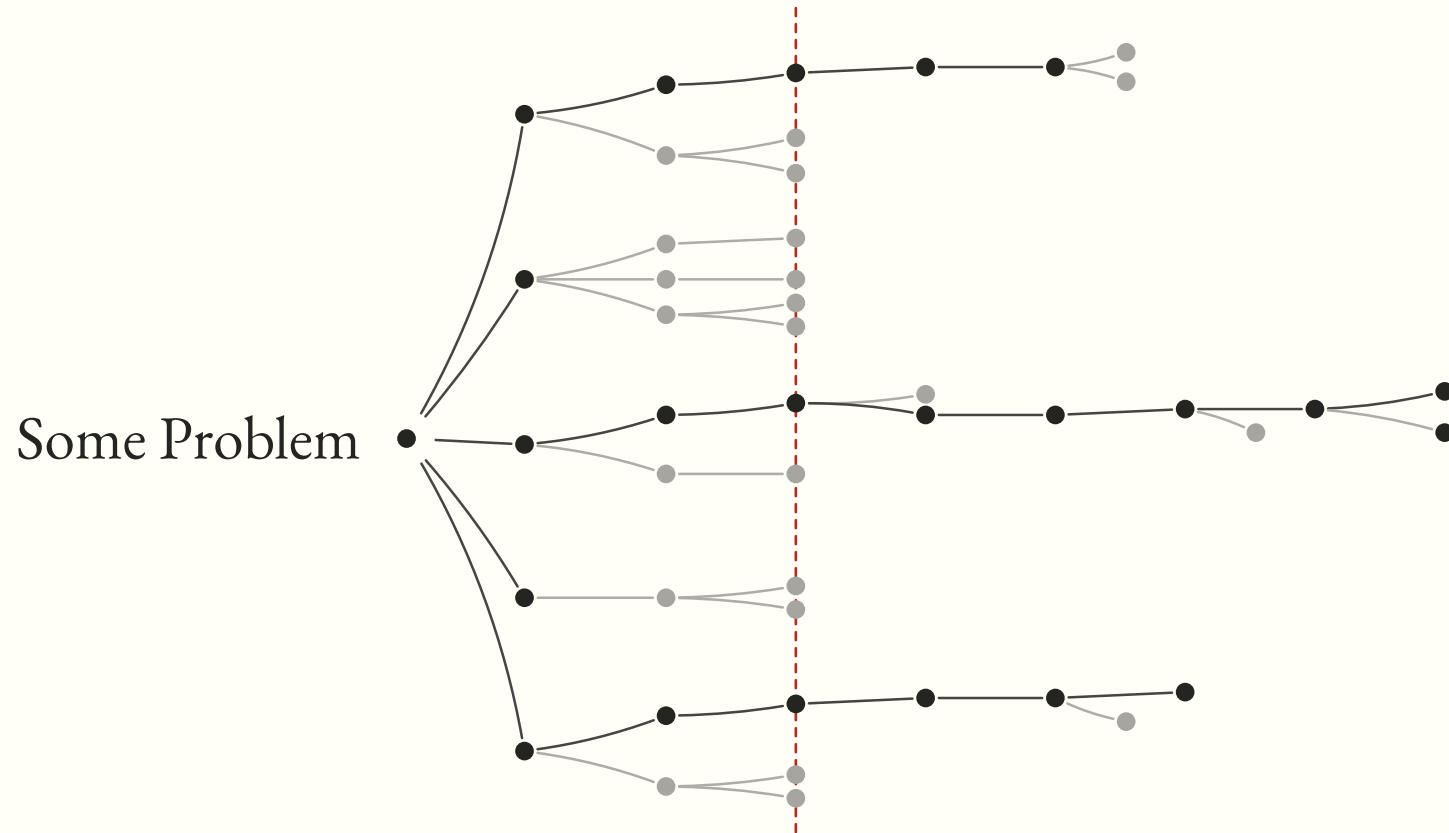
Ideation as a Tree



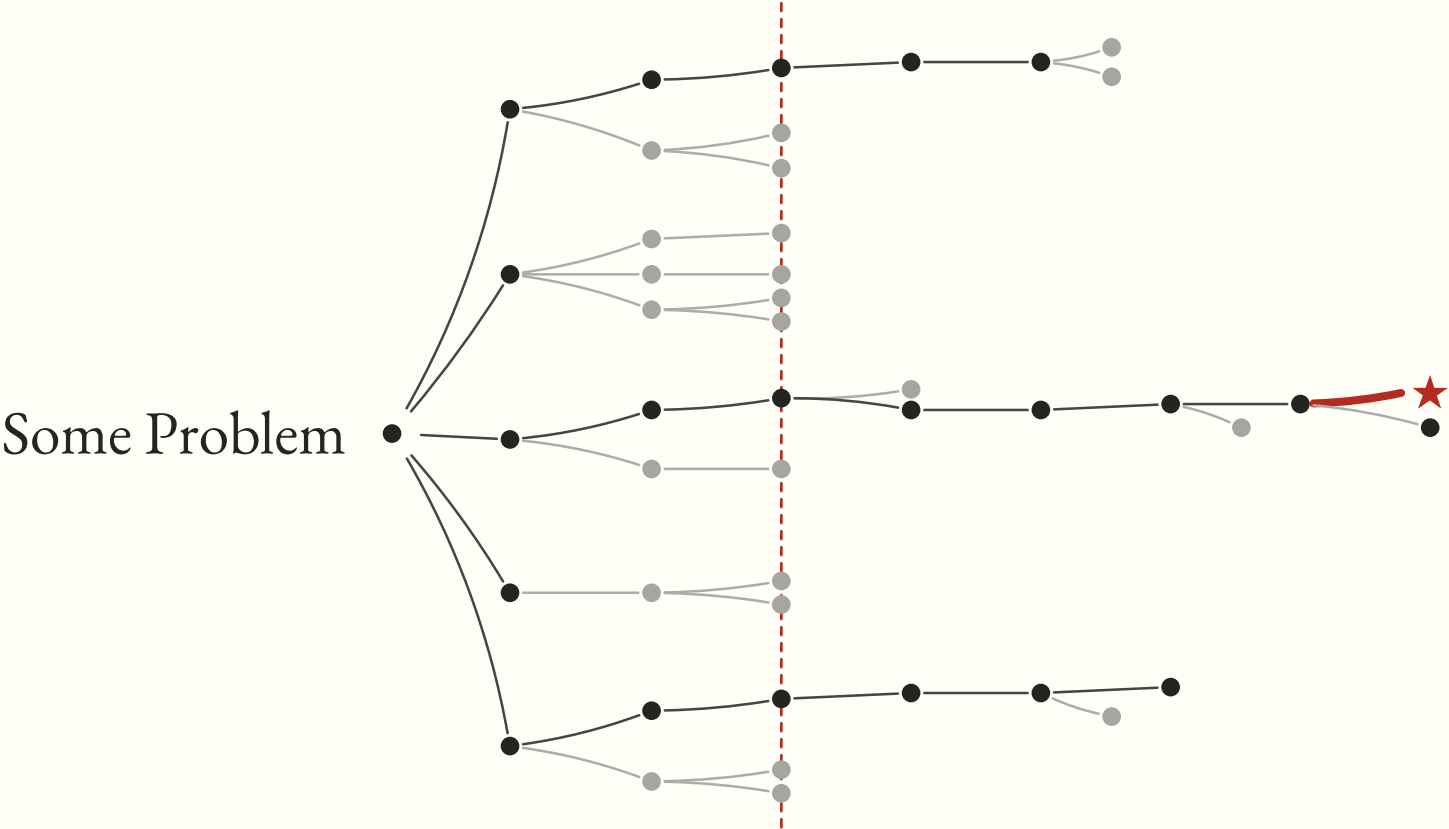
Ideation as a Tree



Ideation as a Tree



Ideation as a Tree

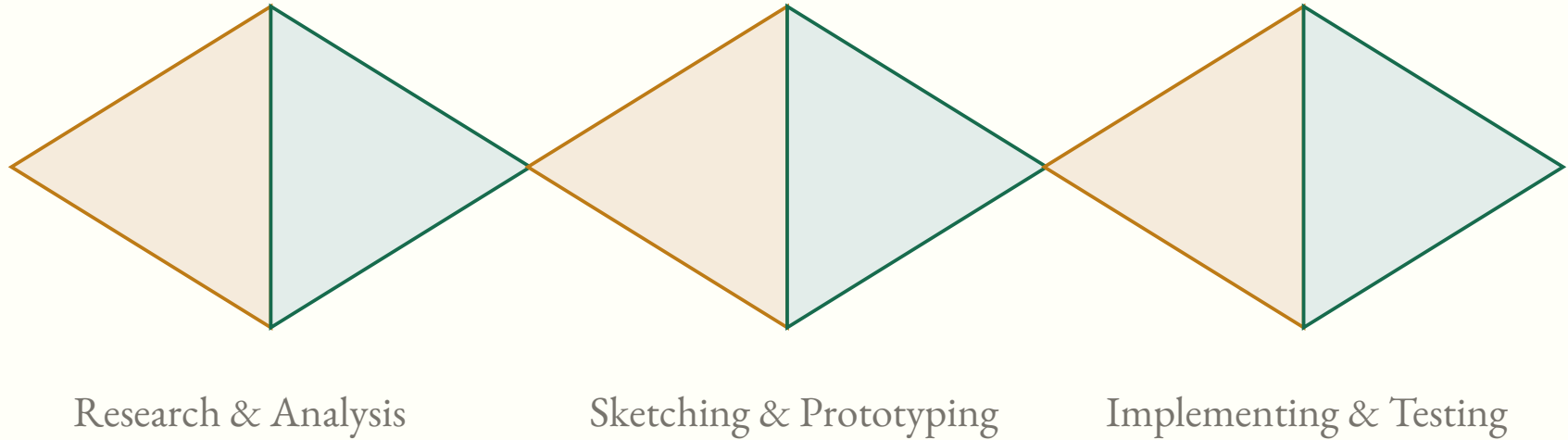


| *The best way to have a good idea is to **have lots of ideas**.*

— *Linus Pauling*

Professor of Chemistry, Caltech · UC San Diego · Stanford — only person awarded two unshared Nobel Prizes

Elaborate and Reduce



Sketching is fundamental to ideation and design.

Traditional disciplines — industrial design, graphic design, architecture — make extensive use of sketches to develop, explore, communicate, and evaluate ideas.

— *Tobidi, Buxton, Baecker & Sellen*

User Sketches: A Quick, Inexpensive, and Effective Way to Elicit More Reflective User Feedback, NordiCHI 2006

*Sketching has long been a **best practice** for designers. Through sketches, designers follow a generative process of **developing, honing, and choosing ideas**. Designers also use sketches to discuss, exchange, and critique ideas with others.*

— Greenberg, Carpendale, Marquardt & Buxton

Sketching User Experiences: The Workbook, 2012

*Sketches allow for a **dialog between the sketch and the viewer** — even when the viewer is the sketcher themselves — that **facilitates better understanding** of the problem, and in turn, generation of new ideas.*

— Tohidi, Buxton, Baecker & Sellen

NordiCHI 2006

*Sketching is **not about drawing!***

Often the best sketches are just rough line drawings.

Sketching is a Conversation with the Self

